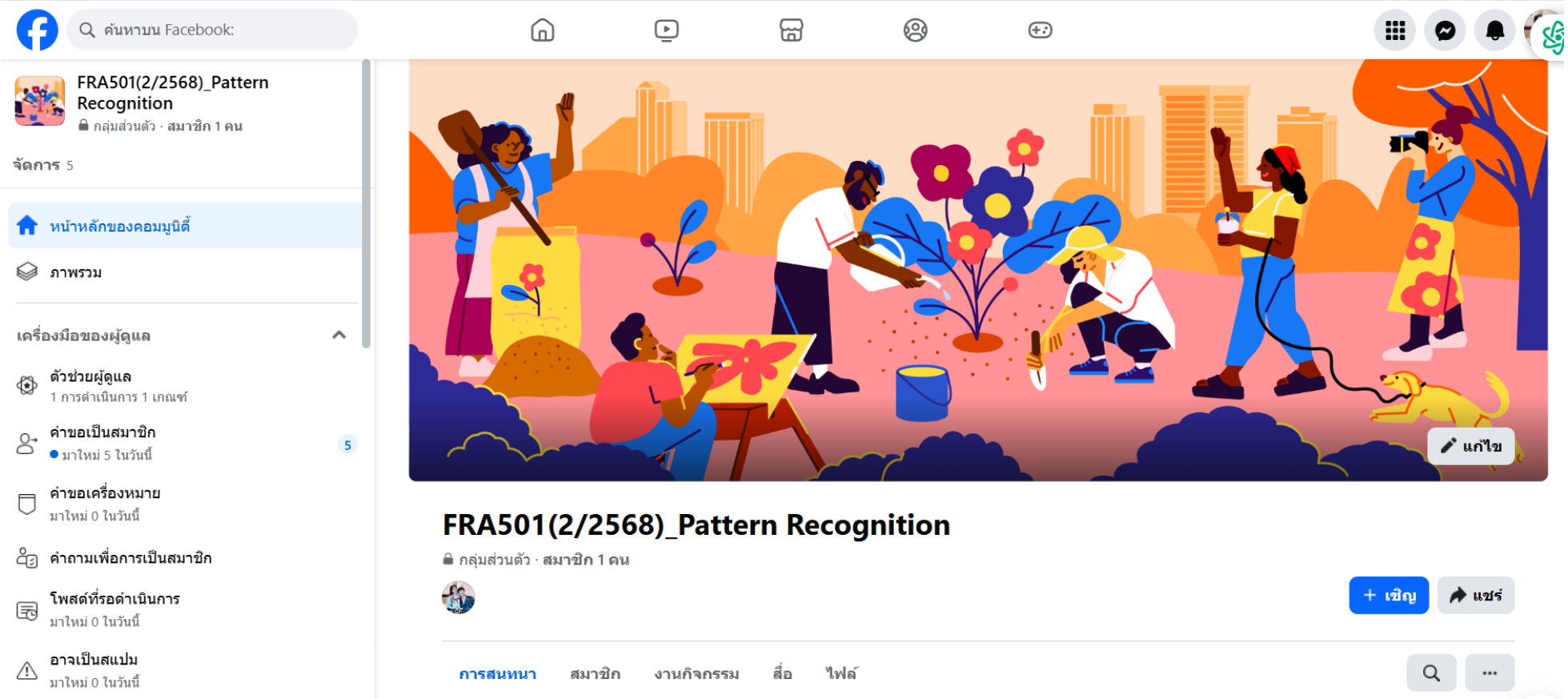
INTRODUCTION

FRA501 Pattern Recognition

Information

Facebook

https://web.facebook.com/share/g/1Jo3sD4Yvf/



Syllabus

Week	Date	Lead Lecturer	Topic	Activities
1	13/01/2026	Aj Paisit	Introduction to Pattern Recognition	
2	20/01/2026	Aj Paisit	Break	
3	27/01/2026	Aj Paisit	K-mean and Regression	**HW1(25%) assigned**
4	3/2/2026	Aj Paisit	MLE (Maximum likelihood estimate), MAP (Maximum A Posteriori)	
5	10/2/2026	Aj Paisit	Bayes Rule and Naive Bayes	**HW2(25%) assigned**
Exam1	17/02/2026	Aj Paisit	Break (No exam for this class)	**HW1 submitted**
6	24/02/2026	Aj Paisit	GMMs (Gaussian mixture models) and EM (Expectation maximization algorithm)	
7	3/3/2026	Aj Paisit	Dimensionality Reduction and Visualization Techniques Part01	
8	10/3/2026	Aj Paisit	Dimensionality Reduction and Visualization Techniques Part02	**HW2 submitted**
9	17/03/2026	Aj Paisit	Support Vector Machine (SVM)	
Exam2 (10)	24/03/2026	Aj Paisit	Break (No exam for this class)	
Exam2	31/03/2026	Aj Paisit	Break (No exam for this class)	
Break	7/4/2026	Aj Paisit	Break	
11	14/04/2026	Aj Paisit	Basic CNN for computer vision and Basic RNN for sequential data , optimizers (SGD, learning rate setting, momentum, Nesterov momentum and adaptive learning rate)	**HW3(25%) assigned**
12	21/04/2026	Aj Paisit	RNN and CNN in advance (DL architectures (LeNet, AlexNet, ZFNet, VGG, InceptionV1-3, ResNet, Xception, DenseNet, NasNet, EfficientNet, Modern Convolution block))	
13	28/04/2026	Aj Paisit	Basic NLP, Different recurrent architectures (many-to-many, many-to-one, one-to-many, many-to-many (encoder-decoder)), attention, transformers, pre-trained transformer types, Vision transformer	**HW4(25%) assigned**
14	5/5/2026	Aj Paisit	Unsupervised, Semi-Supervised, Self-Supervised, Contrastive Learning	
15	12/5/2026	Aj Paisit	Deep Generative Models (VAE, Diffusion)	**HW3 submitted**
Exam3	19/05/2026	Aj Paisit	Break (No exam for this class)	
Exam3	26/05/2026	Aj Paisit	Break (No exam for this class)	**HW4 submitted**

Plagiarism Policy

- You are prohibited from sharing your code or solutions with others.
- Any form of plagiarism or copying will result in a zero score for all involved parties.
- Although implementations of these algorithms can be found online, directly copying or pasting such code is strictly not allowed.
- AI policy: Generative AI tools may be used as assistance; however, you are fully accountable for your submitted work. All content must be carefully reviewed and validated by you. Incorrect citations or evident copy-and-paste outputs (e.g., phrases like "Here's a solution to your…") will lead to a zero grade for the entire assignment or project.

Plagiarism vs. Cheating

What is the difference?

Grades

Grading (100%)

- Assignments (25%): 4 times, <= 3 people

> 85% A

>= 80% B+

>= 75% B

>= 70% C+

>= 65% C

>= 60% D+

>= 55% D

< 55% F

The scoring criteria mentioned above are preliminary. The instructor reserves the right to adjust them as deemed appropriate.

Zoom link:

https://kmutt-ac-th.zoom.us/j/94689943574?pwd=cnpYNThWWVNDSTk3bENSNUFHaHFqdz09

Meeting ID: 946 8994 3574

Passcode: 572369

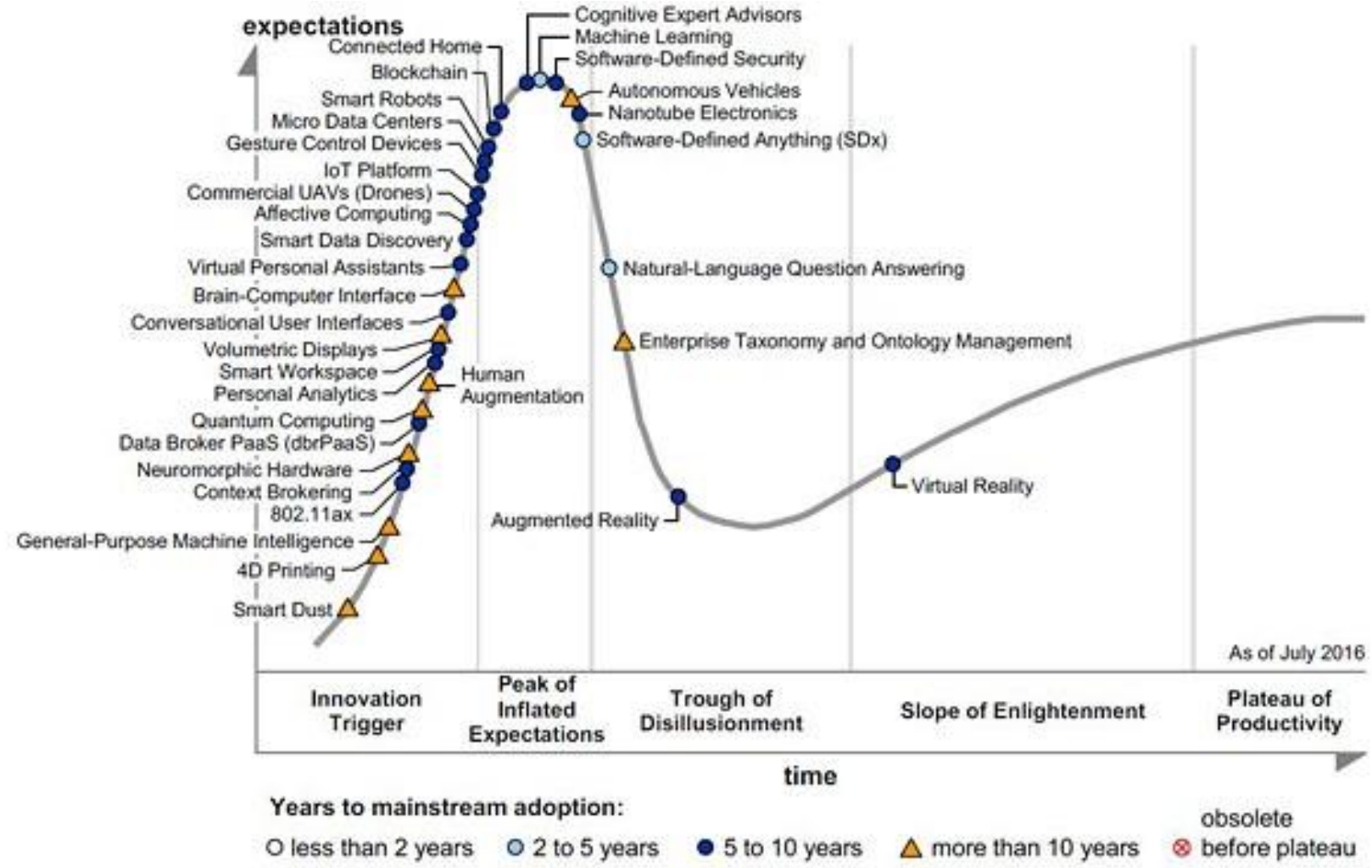
Additional materials

Special Thank!!!
Ekapol Chuangsuwanich, Chulalongkorn University
https://scholar.google.com/citations?hl=en&user=ST-jPeYAAAAJ

Books
1. Richard O. Duda, Peter E. Hart, David G. Stork, Pattern Classification, John Wiley & Sons, 2012.
2. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016.

The machine learning trend 2016

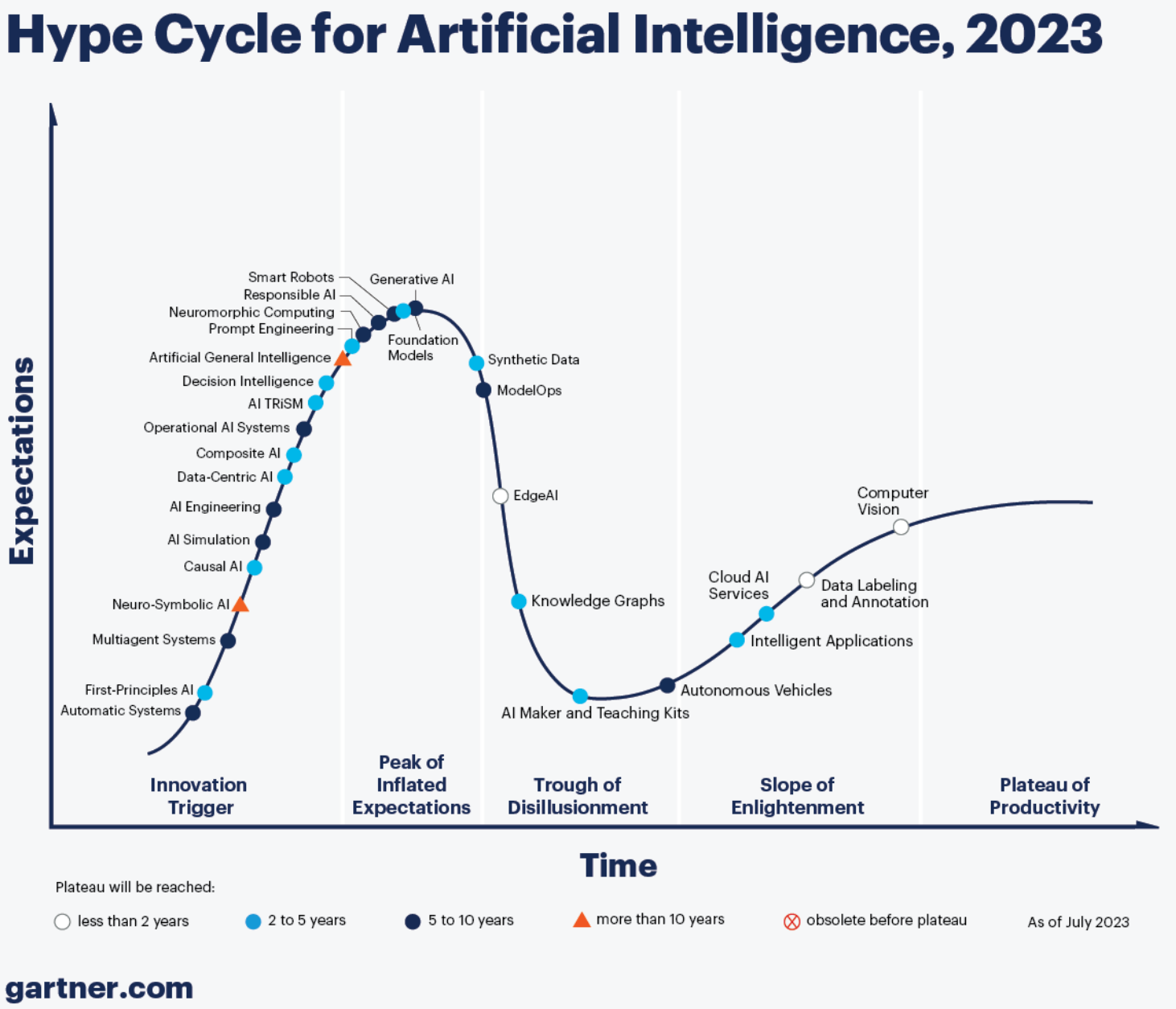
http://www.gartner.com/newsroom/id/3412017



Source: Gartner (July 2016)

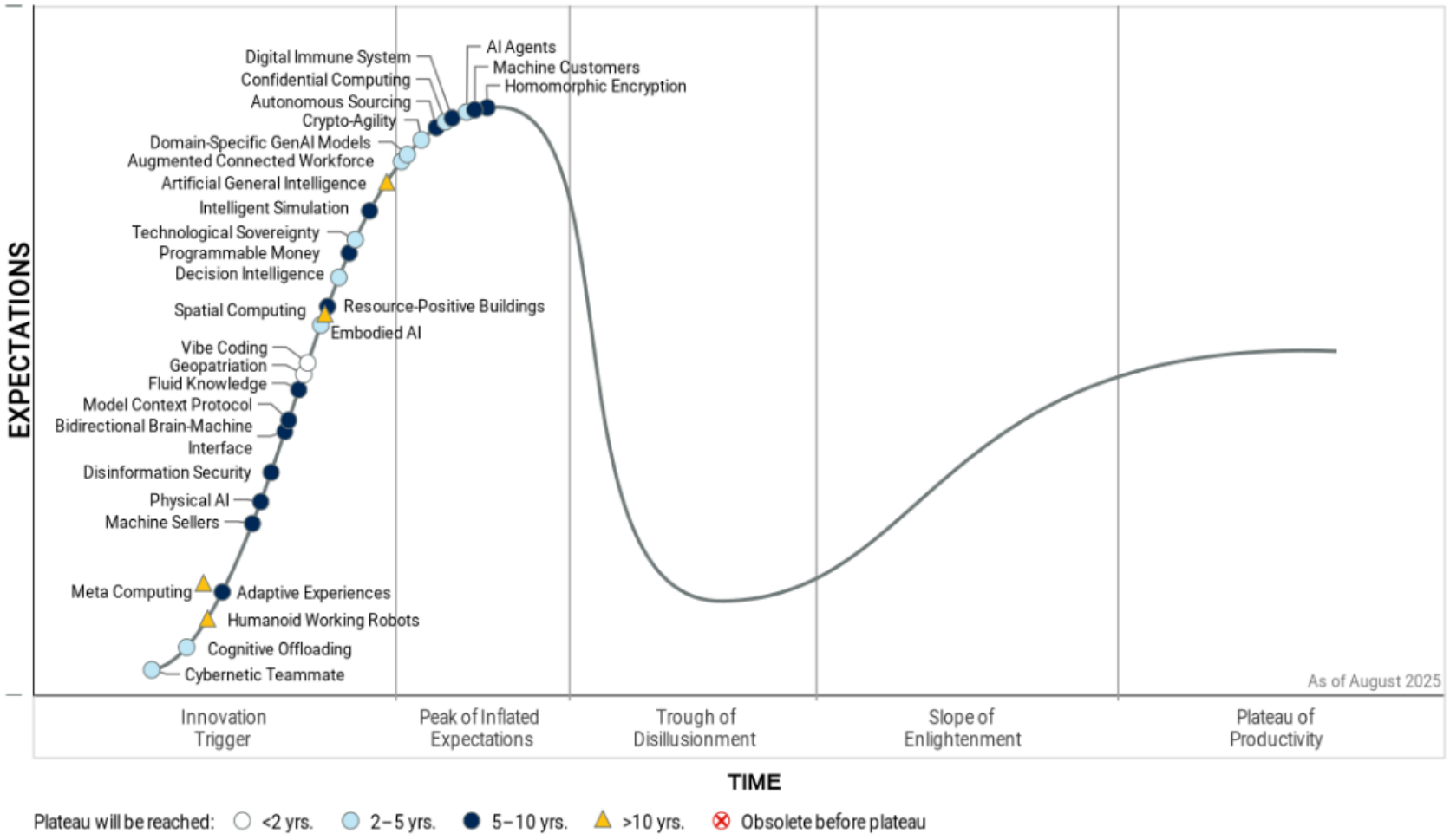
The machine learning trend 2023

- https://www.gartner.com/en/articles/what-s-new-in-artificial-intelligence-from-the-2023-gartner-hype-cycle



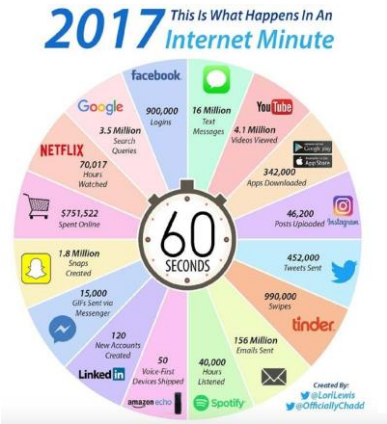
The machine learning trend 2025

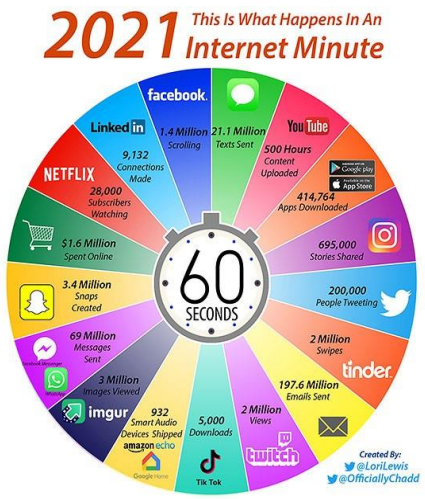
- https://www.gartner.com/en/newsroom/press-releases/2025-09-10-gartner-unveils-top-emerging-technologies-to-support-autonomous-business

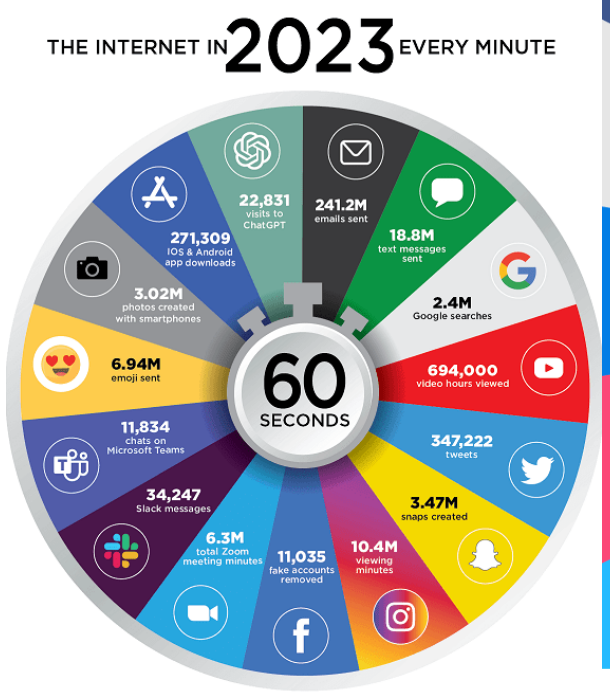


Gartner

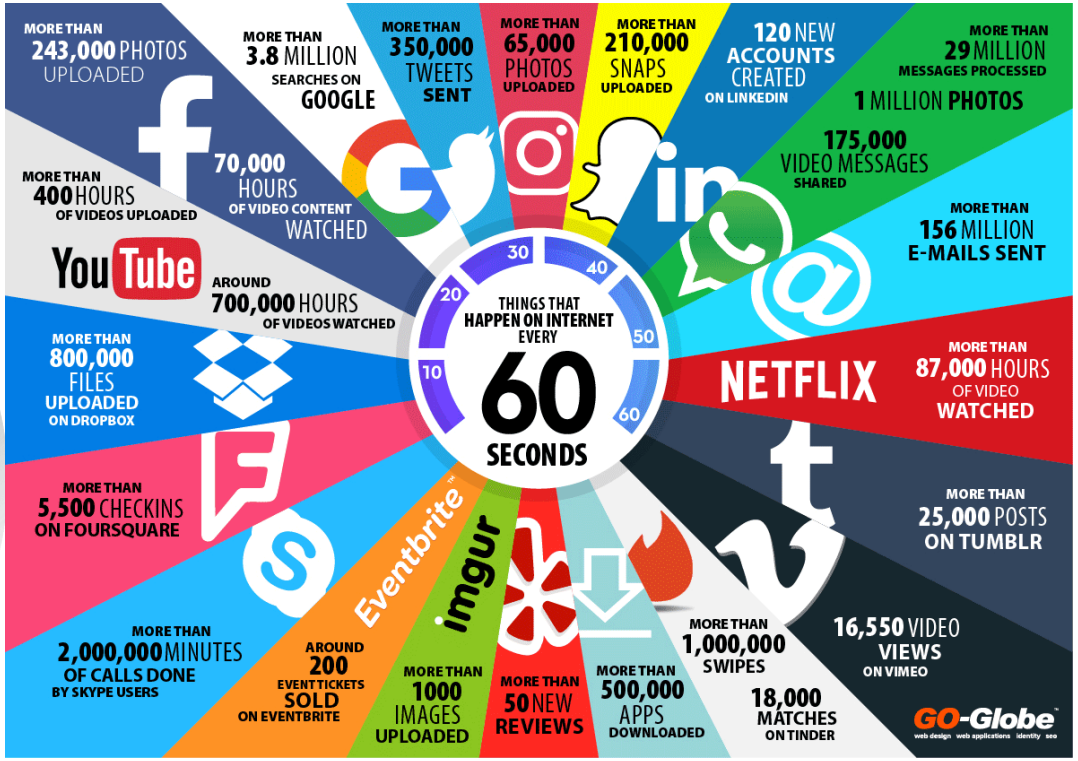
The data era





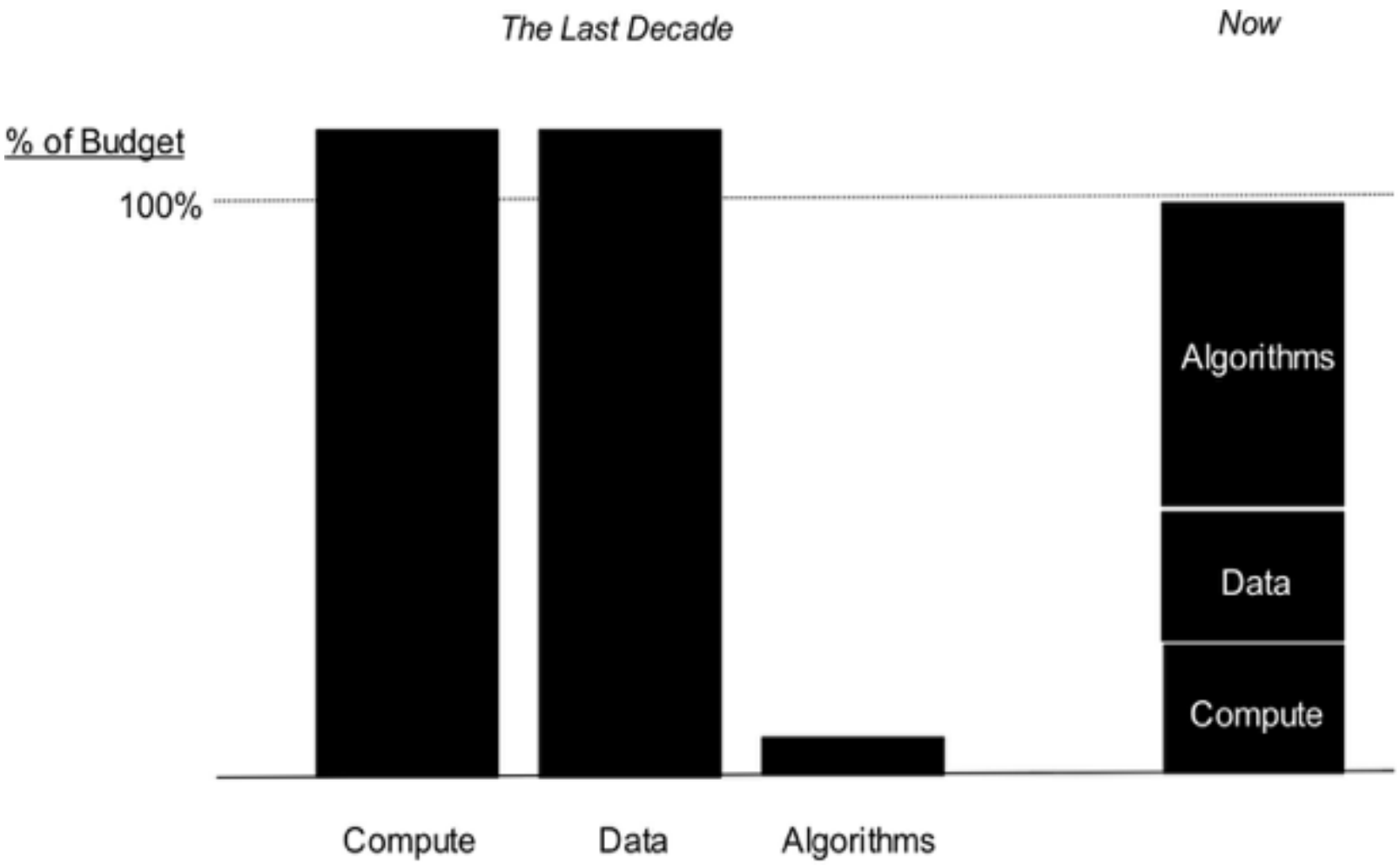


Created by: eDiscovery Today & LTMG



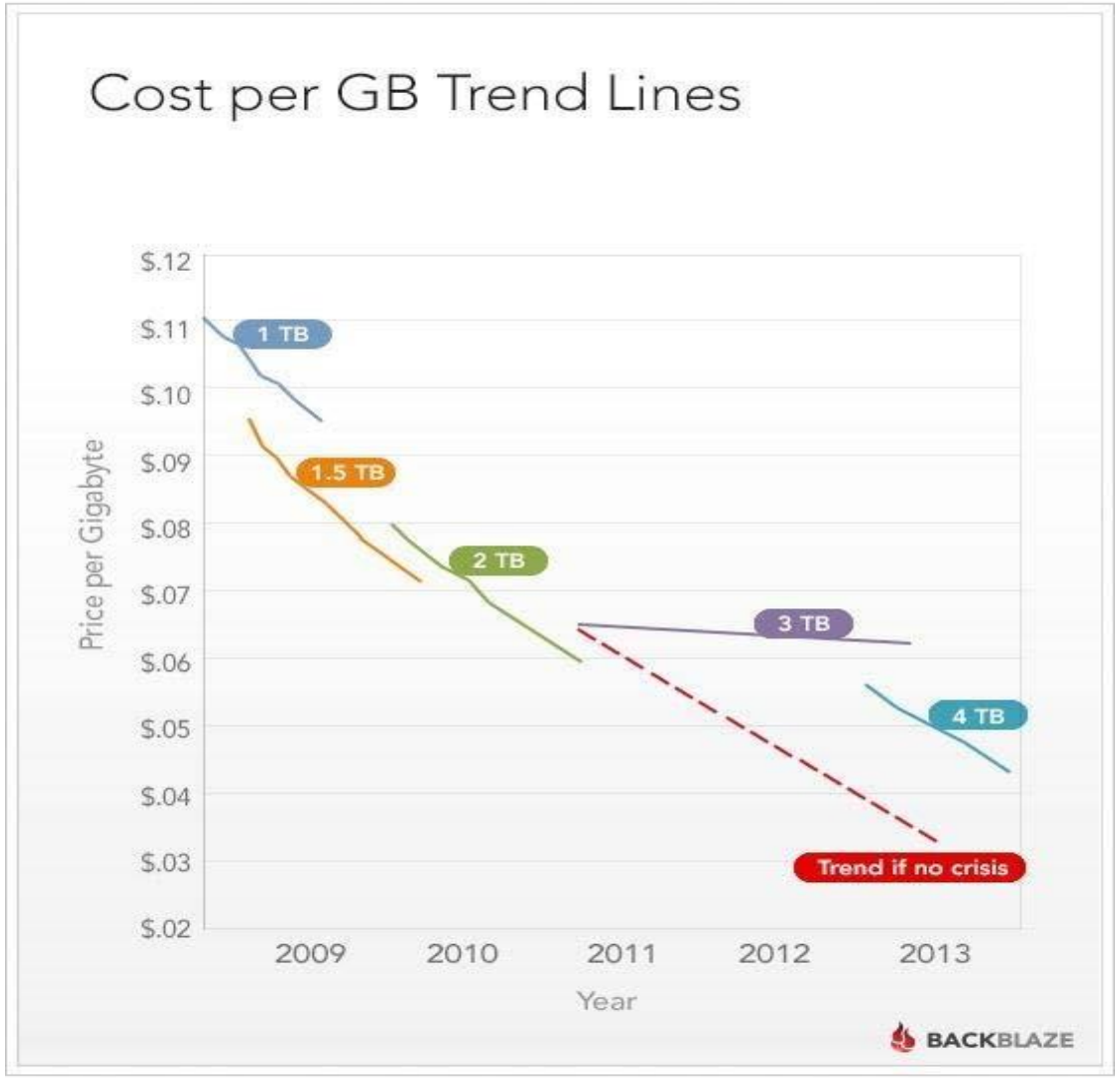
Factors for ML

- Data
- Compute
- Algo

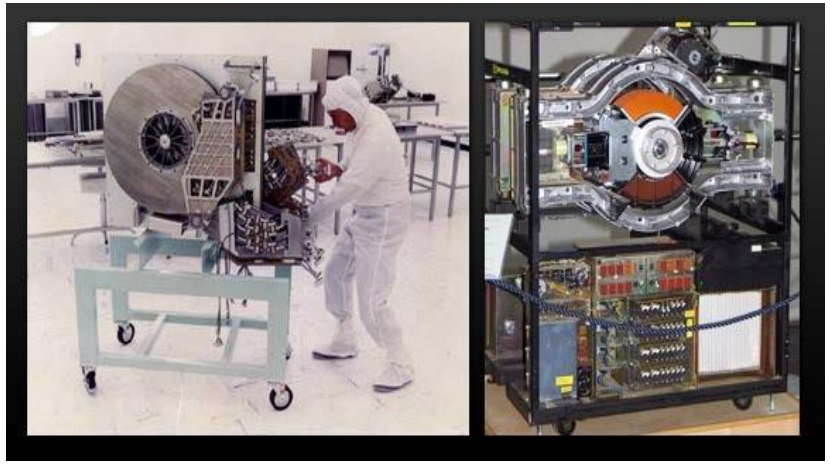


http://www.kdnuggets.com/2017/06/practical-guide-machine-learning-understand-differentiate-apply.html

The cost of storage

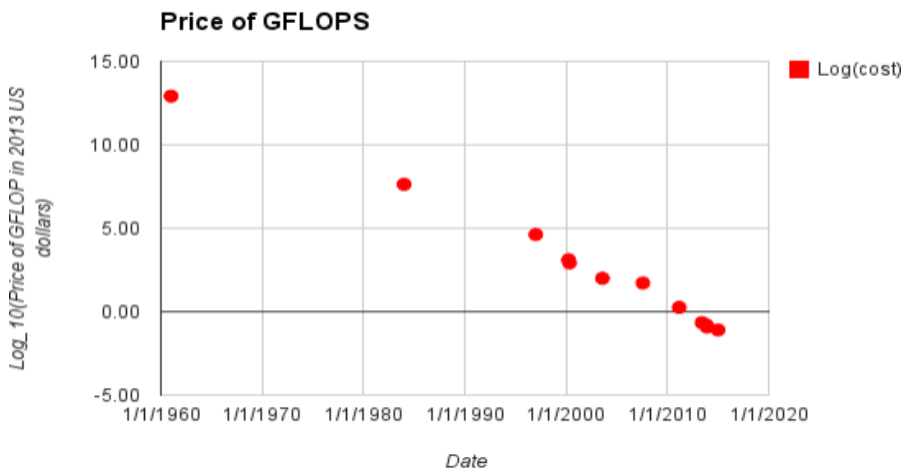


https://www.backblaze.com/blog/farming-hard-drives-2-years-and-1m-later/



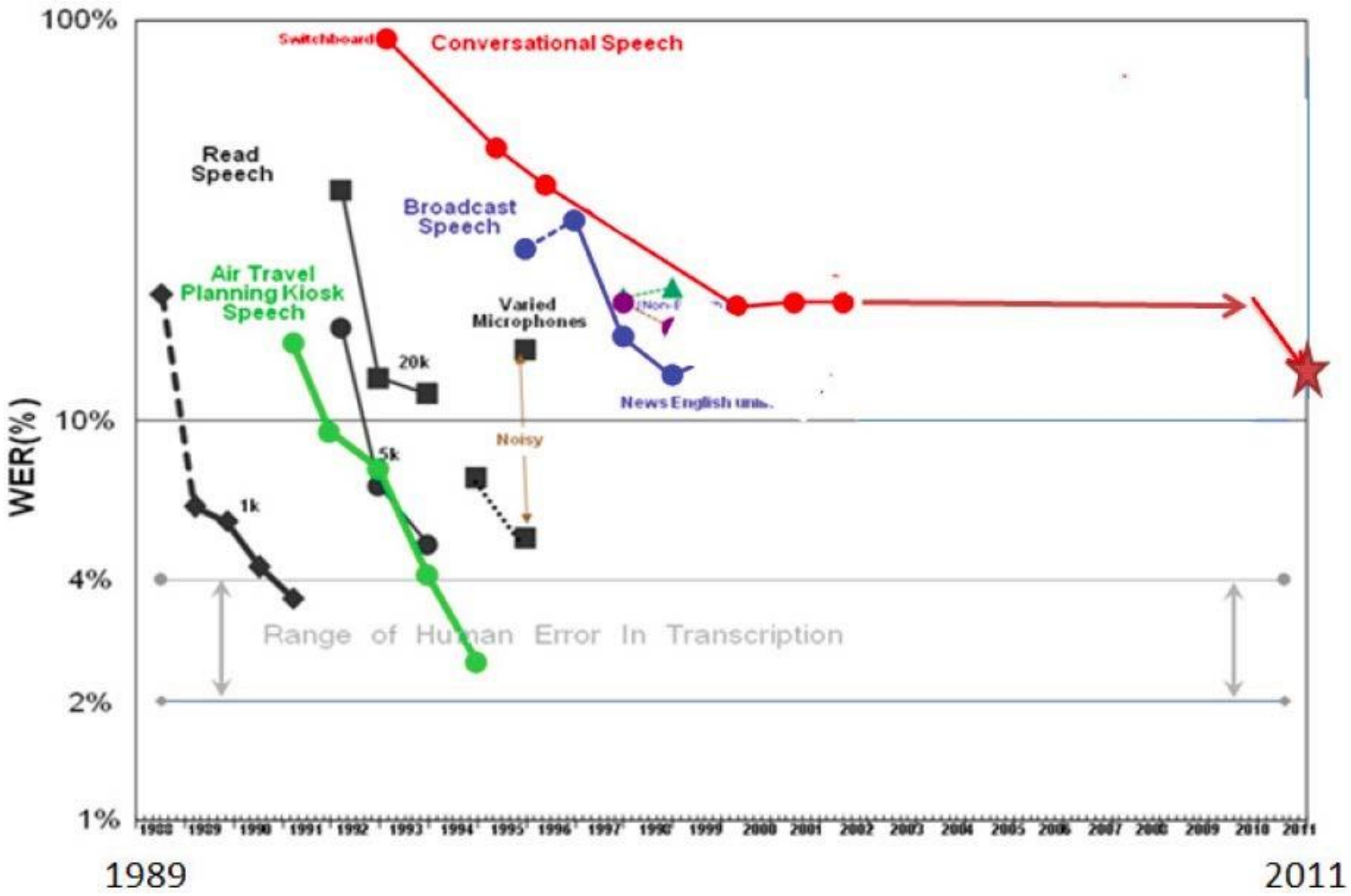
http://royal.pingdom.com/2008/04/08/the-history-of-computer-data-storage-in-pictures/

1980 250MB hard disk drive
250 kg 100k USD (300k USD in today's dollar)



http://aiimpacts.org/trends-in-the-cost-of-computing/

Hitting the sweet spot on performance



http://recognize-speech.com/acoustic-model/knn/benchmarks-comparison-of-different-architectures

What is Pattern Recognition?

- "Pattern recognition is a branch of machine learning that focuses on the recognition of patterns and regularities in data, <u>although it is in some cases considered to be nearly synonymous with machine learning</u>."

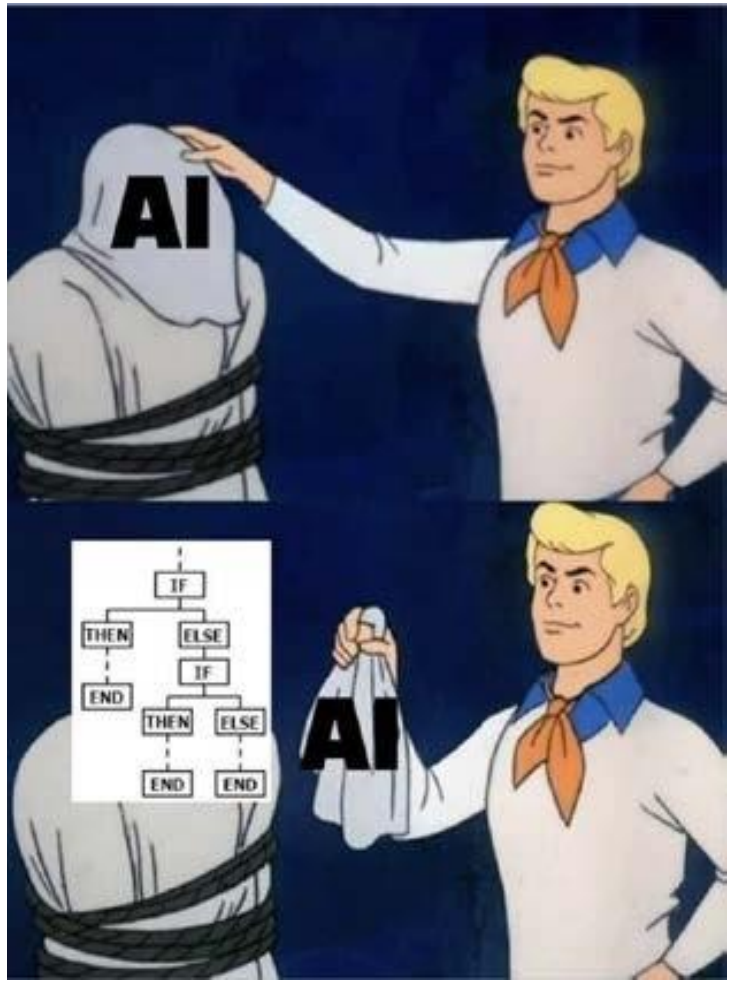
wikipedia

- What about
  - AI
  - Data mining
  - Knowledge Discovery in Databases (KDD)
  - Statistics
  - Data science

What is AI?

- Classical definition
  - A system that <u>appears</u> intelligent
- Populace definition

- Probably what the field means right now
  - ML
    - narrow AI
      - Specialized



https://techsauce.co/pr-news/tcas-use-cloud-computing-and-ai-for-admission

Artificial General Intelligence (AGI)

- "hypothetical ability of an intelligent agent to understand or learn any intellectual task that a human being can."

Wikipedia

Can continue to learn new skills on its own.

Probably not of much interest besides philosophical debates
Works done in baby steps

ML vs PR vs DM vs KDD

- “The short answer is: None. They are … concerned with the same question: how do we learn from data?”

Larry Wasserman – Carnegie Mellon University (CMU) Professor

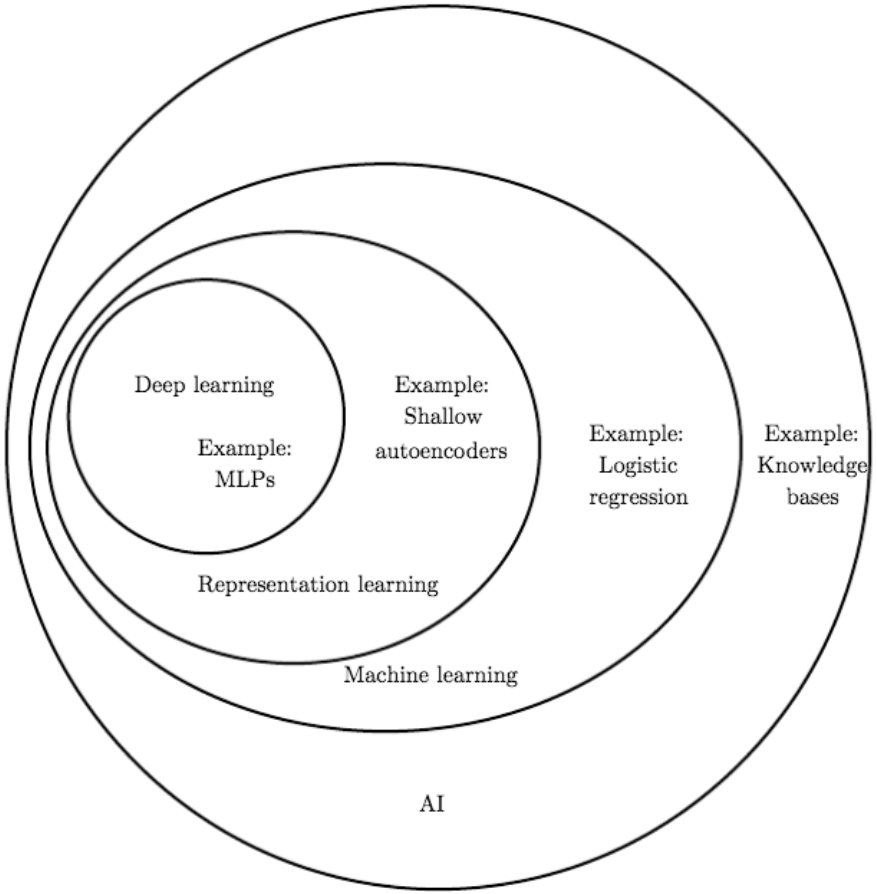
- Nearly identical tools and subject matter

History

- Pattern Recognition started from the engineering community (mainly Electrical Engineering and Computer Vision)
- Machine learning comes out of AI and mostly considered a Computer Science subject
- Data mining starts from the database community

Distinguishing things

- DM – Data warehouse, ETL
- AI – search, swarm intelligence
- PR – Signal processing (feature engineering)



http://www.deeplearningbook.org/

Different terminologies

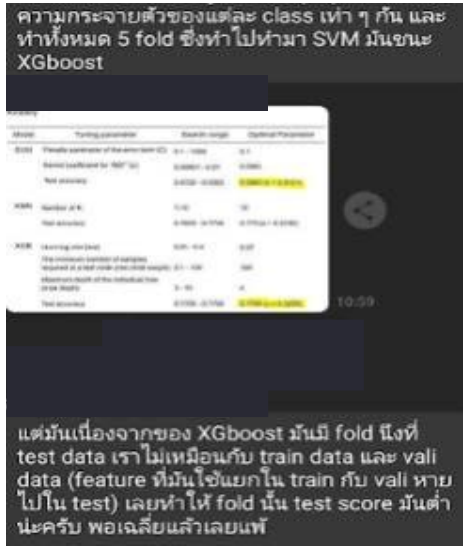
Machine learning	Statistics
network, graphs	model
weights	parameters
learning	fitting
generalization	test set performance
supervised learning	regression/classification
unsupervised learning	density estimation, clustering
large grant = $1,000,000	large grant= $50,000

http://statweb.stanford.edu/~tibs/stat315a/glossary.pdf

Course philosophy

- Going beyond the black box
- In this course you will
  - Understand models on a deeper level
  - Implement stuff from scratch

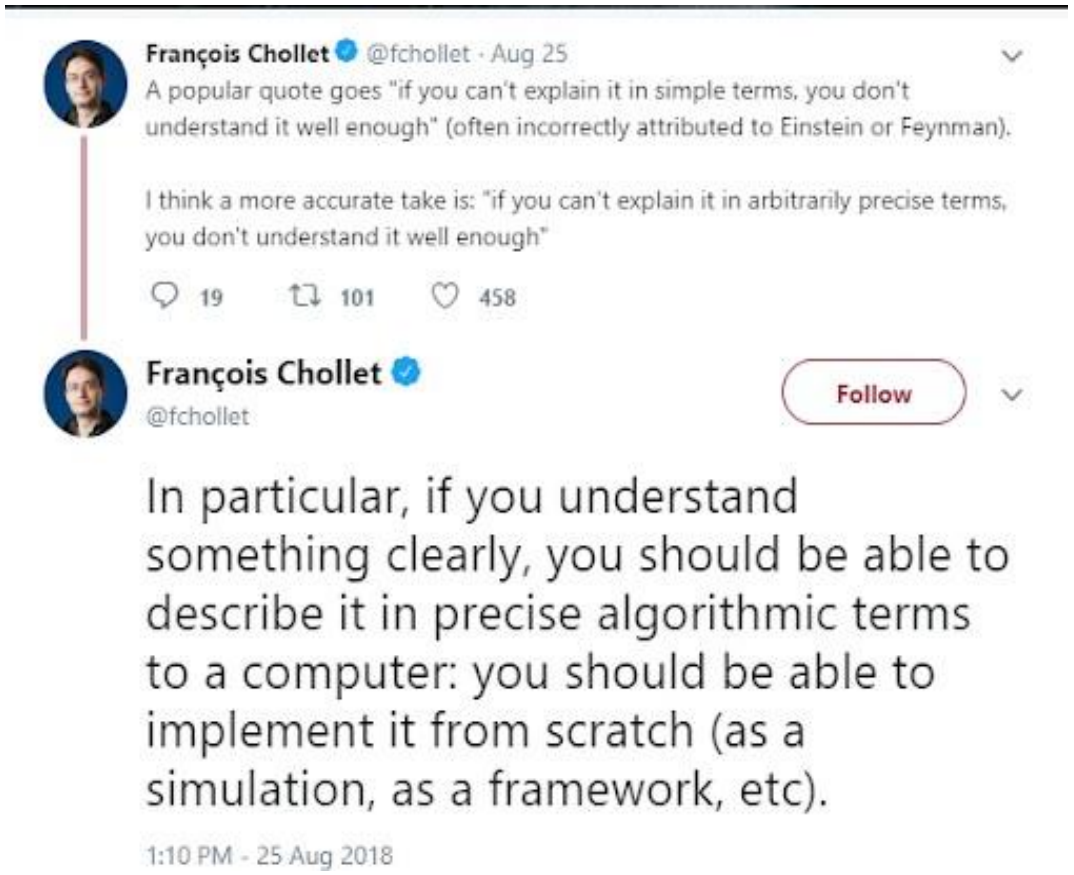
The danger zone



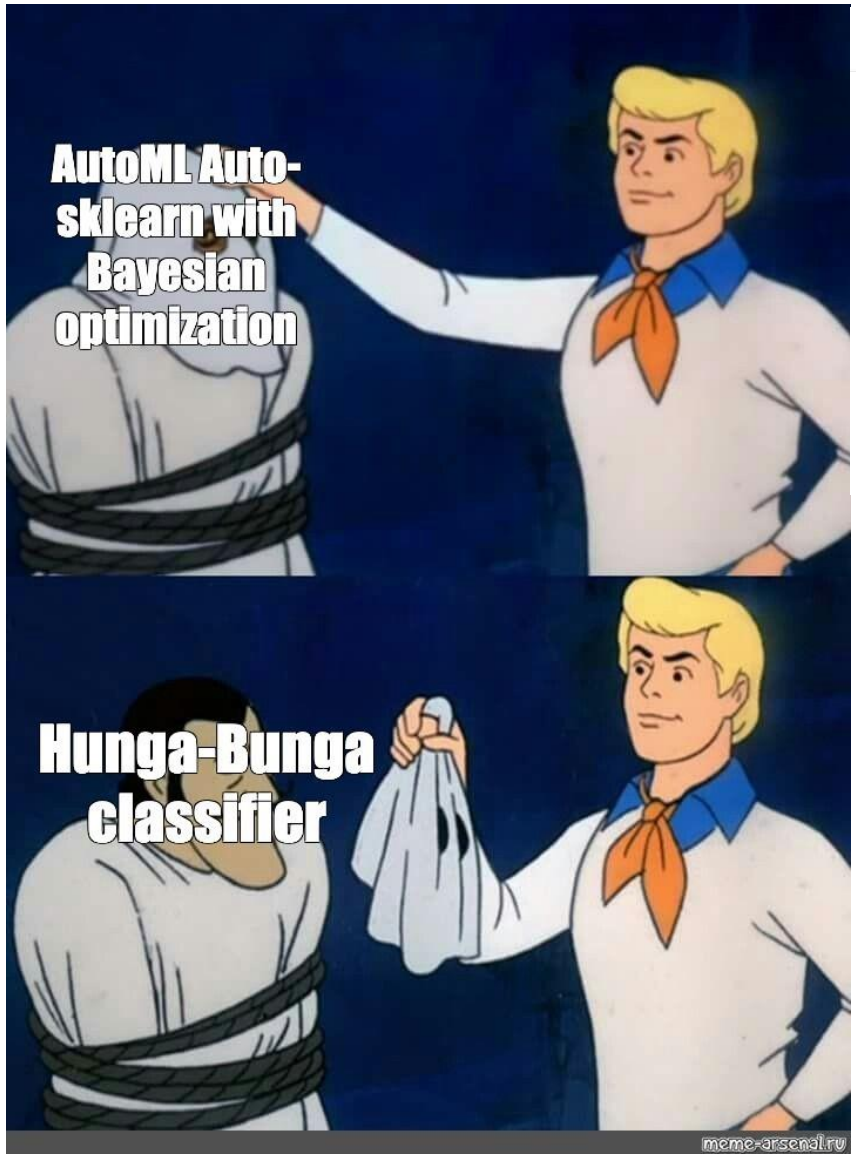
อาจารย์ครับ ผมมีประโยคตัดเป็นคำ ใส่เป็น word bag ที่มีขนาดเท่ากับ vocab
ใส่เป็นความถี่ของคำในประโยค
แล้วผ่าน TF-IDF แล้วเข้า Multinomial NB มันพังกว่าไม่ผ่าน TF-IDF ครับ ผมสงสัยว่าเป็นเพราะอะไร ผมรู้แค่ว่าค่า prob มันเปลี่ยนจริงๆ มันน่าจะดี เพราะผมทำ TF-IDF บน SVM แล้วผลมันดีครับ

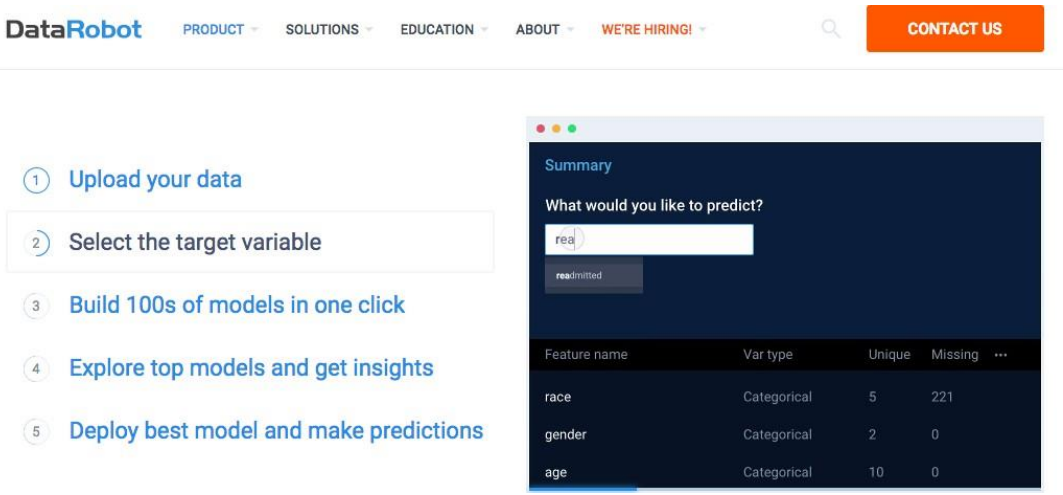
ผมใช้ laplace ด้วยครับ แต่ผลไม่น่าพังขนาดนั้น alpha=1 ครับ

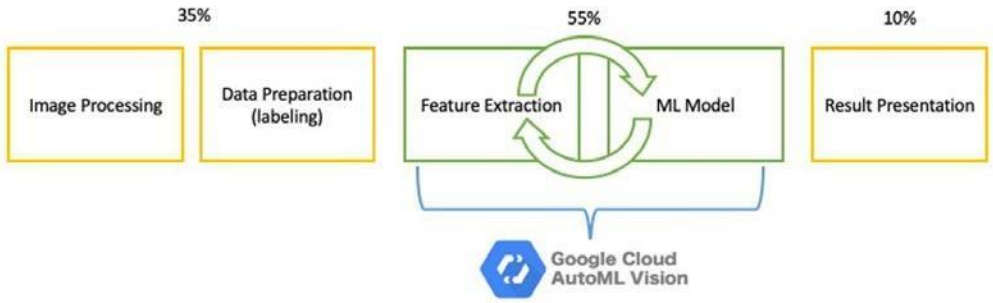
อาจารย์ครับ ควรแบ่ง data ไงดีครับตอนทำ neural net
ผมมี data อยู่ของช่วงวันที่ 9-16 ครับ
คือ ตอนที่ทำ linear กับ pca ผมใช้ train เป็นช่วงวันที่ 9 - 13 ครับ ส่วน test ผมใช้ช่วง 14-15
ตอนทำ neural net ถ้าผมใช้เป็นวันที่ 16 วันเดียวพอไหมครับ หรือควรแบ่ง data ใหม่ครับ
ตอนนี้ training set ผมมีประมาณ 360000 ครับ ส่วน test set มีประมาณ 150000 ครับ



Be better than autoML







AutoML replaces this stage saving 55% of effort and providing better accuracy

https://towardsdatascience.com/ocr-for-scanned-numbers-using-googles-automl-vision-29d193070c64

Types of machine learning

0. Pre-machine learning: rule-base
1. Supervised learning
2. Unsupervised learning
3. Reinforcement learning

Pre-machine learning: 7-segment display

- **Input:** 7 binary values (0,1) forming a display
- Given **x** = (A, B, C, D, E, F, G)
- **Output:** y, either 0, 1, …, 9 or not a number
- **Task:** write a program (a function F) that maps
  x to y; F(**x**) = y

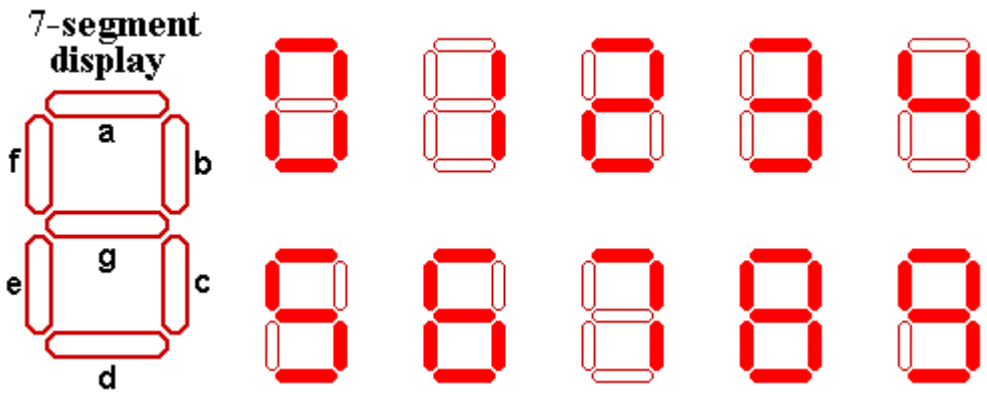
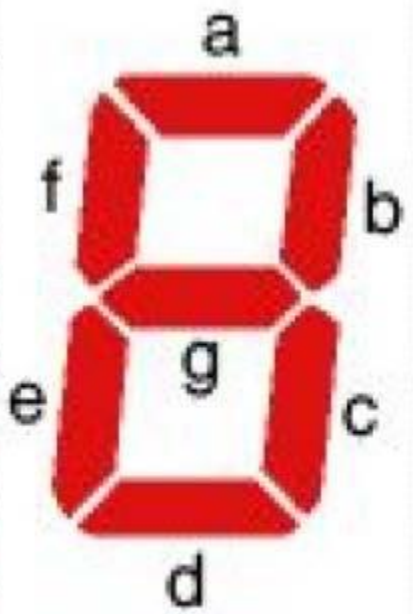


Image from http://www.physics.udel.edu/~watson/scen103/colloq2000/7-seg.html

Mapping function

Y ← X →

Count	a	b	c	d	e	f	g
0	●	●	●	●	●	●	
1		●	●				
2	●	●		●	●		●
3	●	●	●	●			●
4		●	●			●	●
5	●		●	●		●	●
6	●		●	●	●	●	●
7	●	●	●				
8	●	●	●	●	●	●	●
9	●	●	●	●		●	●

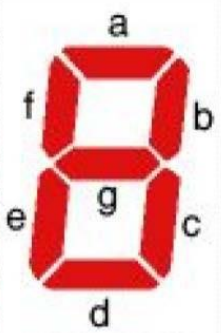


7-segment display

Image from: http://www.instructables.com/id/DIY-7-Segment-Display/

Mapping function

Count	a	b	c	d	e	f	g
0	●	●	●	●	●	●	
1		●	●				
2	●	●		●	●		●
3	●	●	●	●			●
4		●	●			●	●
5	●		●	●		●	●
6	●		●	●	●	●	●
7	●	●	●				
8	●	●	●	●	●	●	●
9	●	●	●	●		●	●



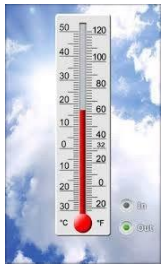
7-segment display

- IF A==1 && B==1 && C==1 && D==1 && E==1 && F==1 && G==0, THEN output(0).
- IF B==1 && C==1, THEN output(1)
- …..
- OTHERWISE, output(“not number”)

F(x)

Learning from data

- Machine learning requires identifying the same ingredients
  - Input, Output, Task



Squire Trelawney, Dr. Livesey, and the rest of those gentlemen having asked me to write down the whole particulars about Treasure Island, from the beginning to the end, keeping nothing back but the bearings of the island, and that only because there is still treasure not yet lifted, I take up my pen in the year of grace 17— and go back to the time when my father kept the Admiral Benbow inn and the brown old seaman with the sabre cut first took up his lodging under our roof.

I remember him as if it were yesterday, as he came ploddling to the inn door, his sea-chest following behind him in a hand-barrow; a tall, strong, heavy, red brown man, his tarry pigtail falling over the shoulder of his soiled blue-coat, his hands ragged and scarred, with black, broken nails, and the sabre cut across one cheek, a dirty, livid white. I remember him looking round the cover and whistling to himself as he did so, and then breaking out in that old sea-song that he sang so often afterwards:

'Fifteen men on the dead man's chest—Yo-ho-ho, and a bottle of rum!' In the high, old tottering voice that seemed to have been tuned and broken at the capstan bars. Then he rapped on the door with a bit of stick like a handspike that he carried, and when my father appeared, called roughly for a glass of rum. This, when it was brought to him, he drank slowly, like a connoisseur, lingering on the taste and still looking about him at the cliffs and up at our signboard.

'This is a handy cove,' says he at length; 'and a pleasant situated grog-shop. Much company, mate?'

My father told him no, very little company, the more was the pity.

'Well, then,' said he, 'this is the berth for me. Here you, messmate,' he cried to the man who trundled the barrow; 'bring up alongside and help up my chest. I'll stay here a bit,' he continued. 'I'm a plain man; rum and bacon and eggs is what I want, and that head up there for to watch ships off. What you mought call me? You mought call me captain. Oh, I see what you're at—there'; and he threw down three or four gold pieces on the threshold. 'You can tell me when I've worked through that,' says he, looking as fierce as a commander.

And indeed had as his clothes were and coarsely as he spoke, he had more of the appearance of a

Real world observations

This is the hardest part of data science and the last part to be replaced by machines.

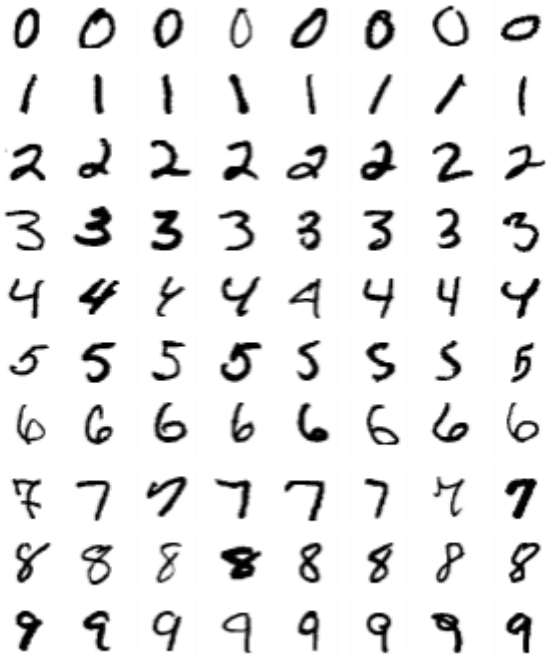
TensorFlow powered cucumber sorter by Makoto Koike
https://youtu.be/4HCE1P-m1l8?si=l0fn1dIYb0Q4o_Zz

https://cloud.google.com/blog/products/gcp/how-a-japanese-cucumber-farmer-is-using-deep-learning-and-tensorflow

An example

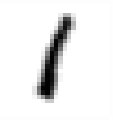
- Handwritten digit recognition
- Input: **x** = 28 x 28 pixel image
- Output: y = digit 0 to 9
- Task: find F(**x**) such that y ≈ F(**x**)

Goal of machine learning is to find the <u>best</u> F(**x**) automatically from data



Supervised learning

- Learn a classifier F from a training set (input-output pairs)
  - $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), (\mathbf{x}_3, y_3), \ldots, (\mathbf{x}_n, y_n)\}$

x	y
	0
	1
	2

Need a training set for training.
Training = finding (optimizing) a good function f

Labeling (i.e., assigning y for each **x** in the training set) is typically done manually.

Types of machine learning

1. Supervised learning

   Learn a model F from pairs of (**x**,y)

2. Unsupervised learning

   Discover the hidden structure in unlabeled data **x** (no y)

3. Reinforcement learning

   Train an agent to take appropriate actions in an environment by maximizing rewards

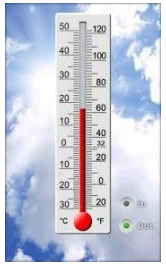
Typical workflow of machine learning

1. Feature extraction (getting the **x**)
2. Modeling
   - Training (getting the function F)
3. Evaluation
   - Metrics (defining what's the best function F)
   - Testing (getting the y for unseen inputs)

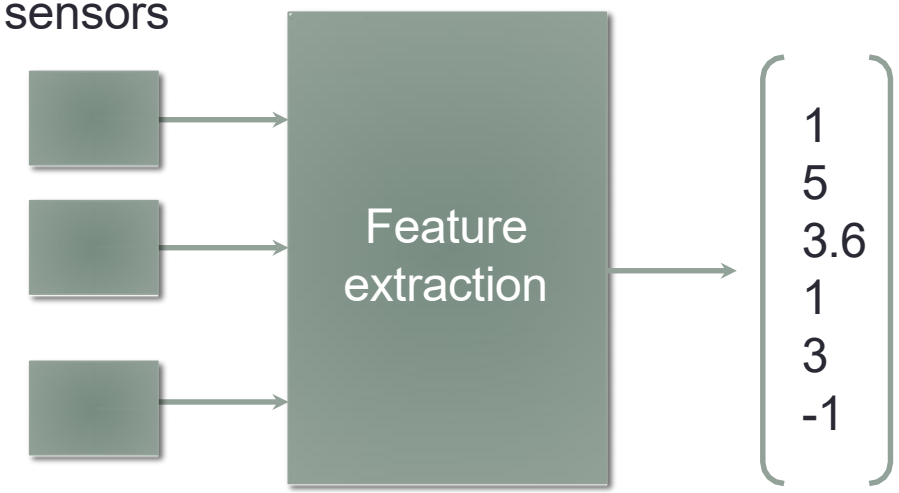
Typical workflow of machine learning

- The typical workflow

Real world observations



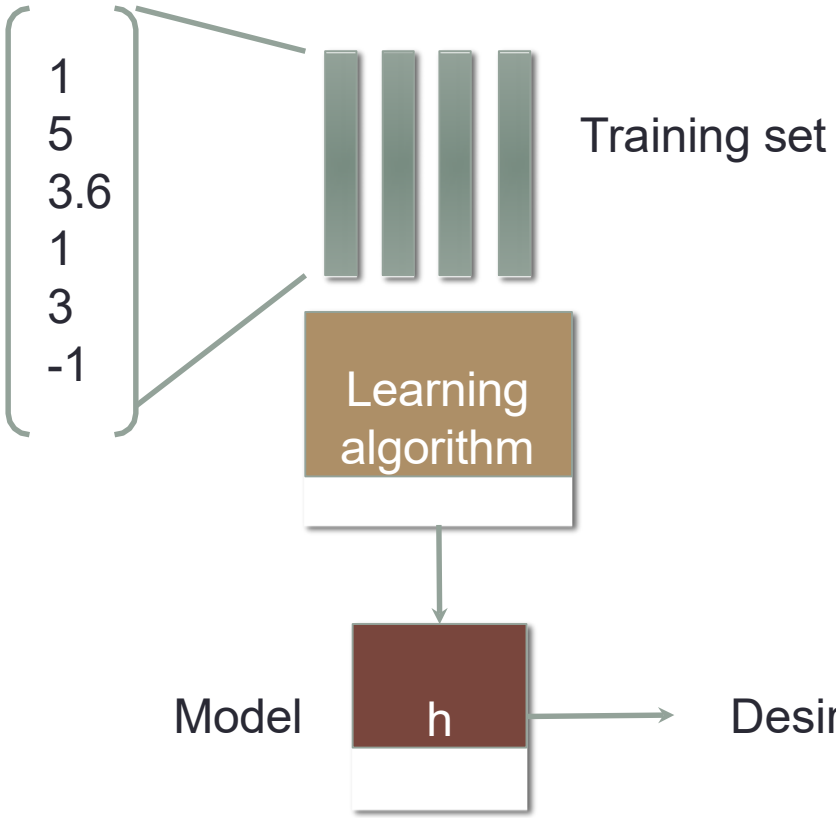
sensors



$$\begin{pmatrix} 1 \\ 5 \\ 3.6 \\ 1 \\ 3 \\ -1 \end{pmatrix}$$

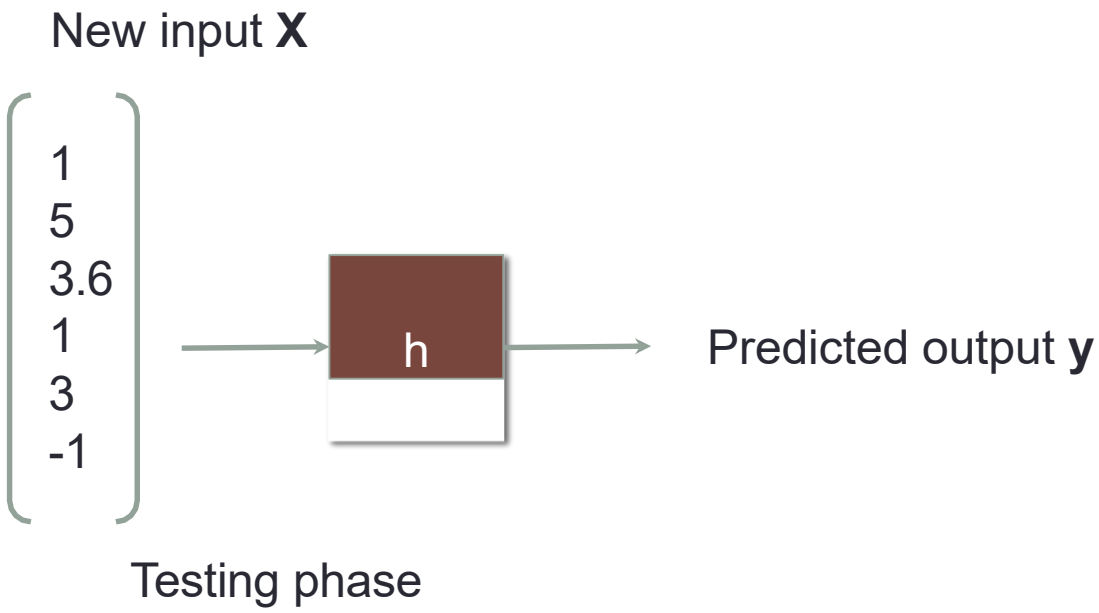
Feature vector **x**

How do we learn from data?



Training phase

How do we learn from data?



Feature extraction

- The process of extracting meaningful information related to the goal
- A distinctive characteristic or quality
- Example features

data1

Squire Trelawney, Dr. Livesey, and the rest of those gentlemen having asked me to write down the whole particulars about Treasure Island, from the beginning to the end, keeping nothing back but the bearings of the island, and that only because there is still treasure not yet lifted, I take up my pen in the year of grace 17— and go back to the time when my father kept the Admiral Benbow inn and the brown old seaman with the sabre cut first took up his lodging under our roof.

I remember him as if it were yesterday, as he came ploddling to the inn door, his sea-chest following behind him in a hand-barrow; a tall, strong, heavy, nut brown man, his tarry pigtail falling over the shoulder of his soiled blue coat, his hands ragged and scarred, with black, broken nails, and the sabre cut across one cheek, a dirty, livid white. I remember him looking round the cover and whistling to himself as he did so, and then breaking out in that old sea-song that he sang so often afterwards:

"Fifteen men on the dead man's chest—Yo-ho-ho, and a bottle of rum!" In the high, old tottering voice that seemed to have been nursed and broken at the capstan bars. Then he rapped on the door with a bit of stick like a handspike that he carried, and when my father appeared, called roughly for a glass of rum. This, when it was brought to him, he drank slowly, like a connoisseur, lingering on the taste and still looking about him at the cliffs and up at our signboard.

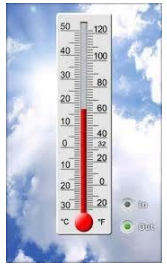
"This is a handy cove," says he at length; "and a pleasant sittyated grog-shop. Much company, mate?"

My father told him no, very little company, the more was the pity.

"Well, then," said he, "this is the berth for me. Here you, mate," he cried to the man who trundled the barrow; "bring up alongside and help up my chest. I'll stay here a bit," he continued. "I'm a plain man; rum and bacon and eggs is what I want, and that head up there for to watch ships off. What you mought call me? You mought call me captain. Oh, I see what you're at—there"; and he threw down three or four gold pieces on the threshold. "You can tell me when I've worked through that," says he, looking as fierce as a commander.

And indeed had as his clothes were and coarsely as he spoke, he had none of the appearance of a

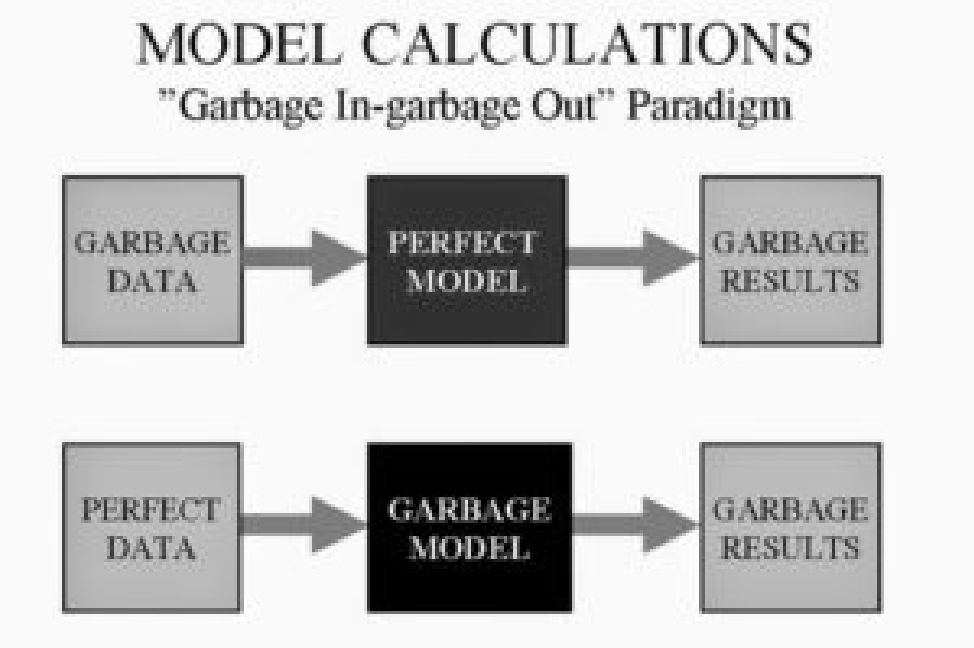
data2

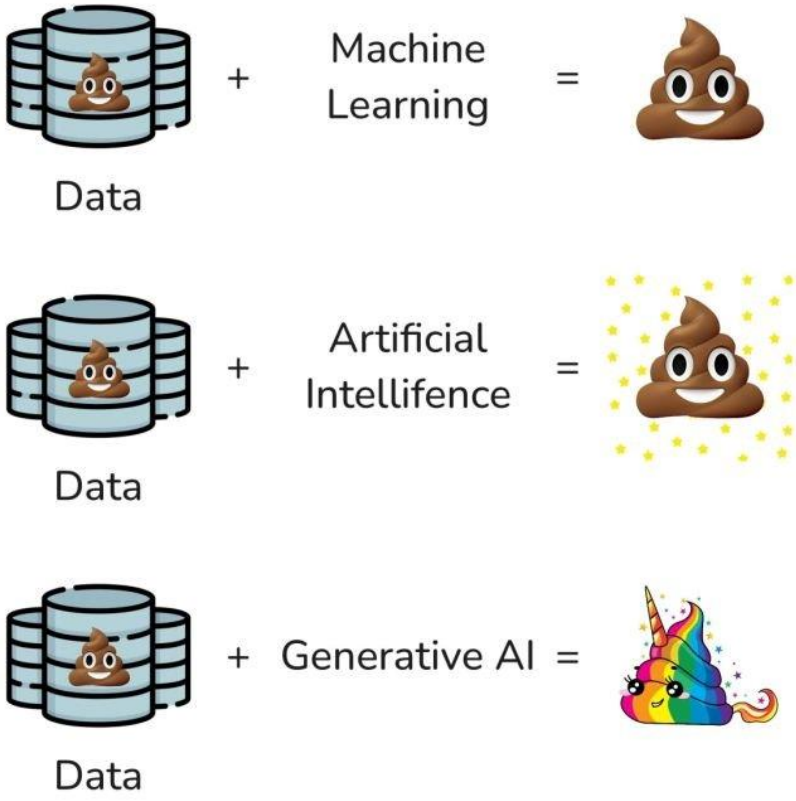


data3

Garbage in Garbage out

- The machine is as intelligent as the data/features we put in
- “Garbage in, Garbage out”
- Data cleaning is often done to reduce unwanted things





https://precisionchiroco.com/garbage-in-garbage-out/

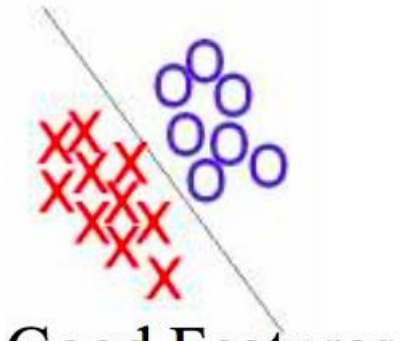
The need for data cleaning

However, good models should be able to handle some dirtiness!

https://www.linkedin.com/pulse/big-data-conundrum-garbage-out-other-challenges-business-platform

Feature properties

- The quality of the feature vector is related to its ability to discriminate samples from different classes

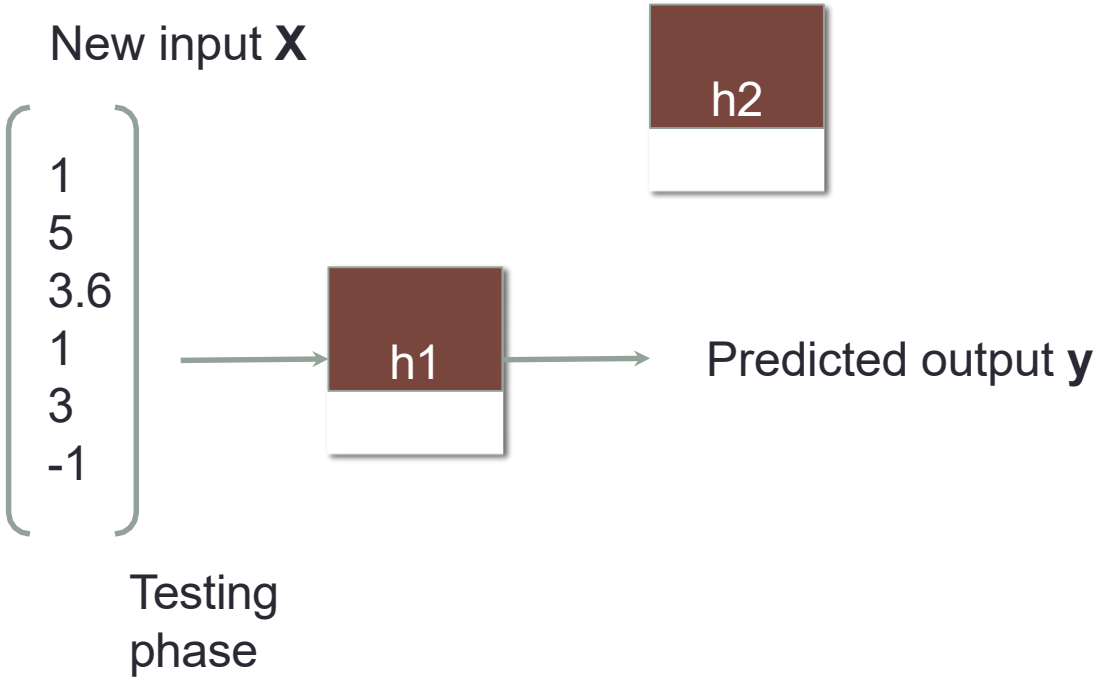


Good Features

Bad Features

Model evaluation

How to compare h1 and h2?



Metrics

- Compare the output of the models
  - Errors/failures, accuracy/success
- We want to quantify the error/accuracy of the models
- How would you measure the error/accuracy of the following

Ground truths

- We usually compare the model predicted answer with the correct answer.
- What if there is no real answer?
  - How would you rate machine translation?

ไปไหน

Model A: Where are you going?
Model B: Where to?

Designing a metric can be tricky, especially when it’s subjective

Ground truths can be hard

Slides from https://github.com/goldmermaid/mlrs

Credit to Andrej Karpathy

Labelling

"Label lane lines"

Credit to Andrej Karpathy

Labelling issues

How do you annotate lane lines when they do this?

abel lane lines"

Credit to Andrej Karpathy

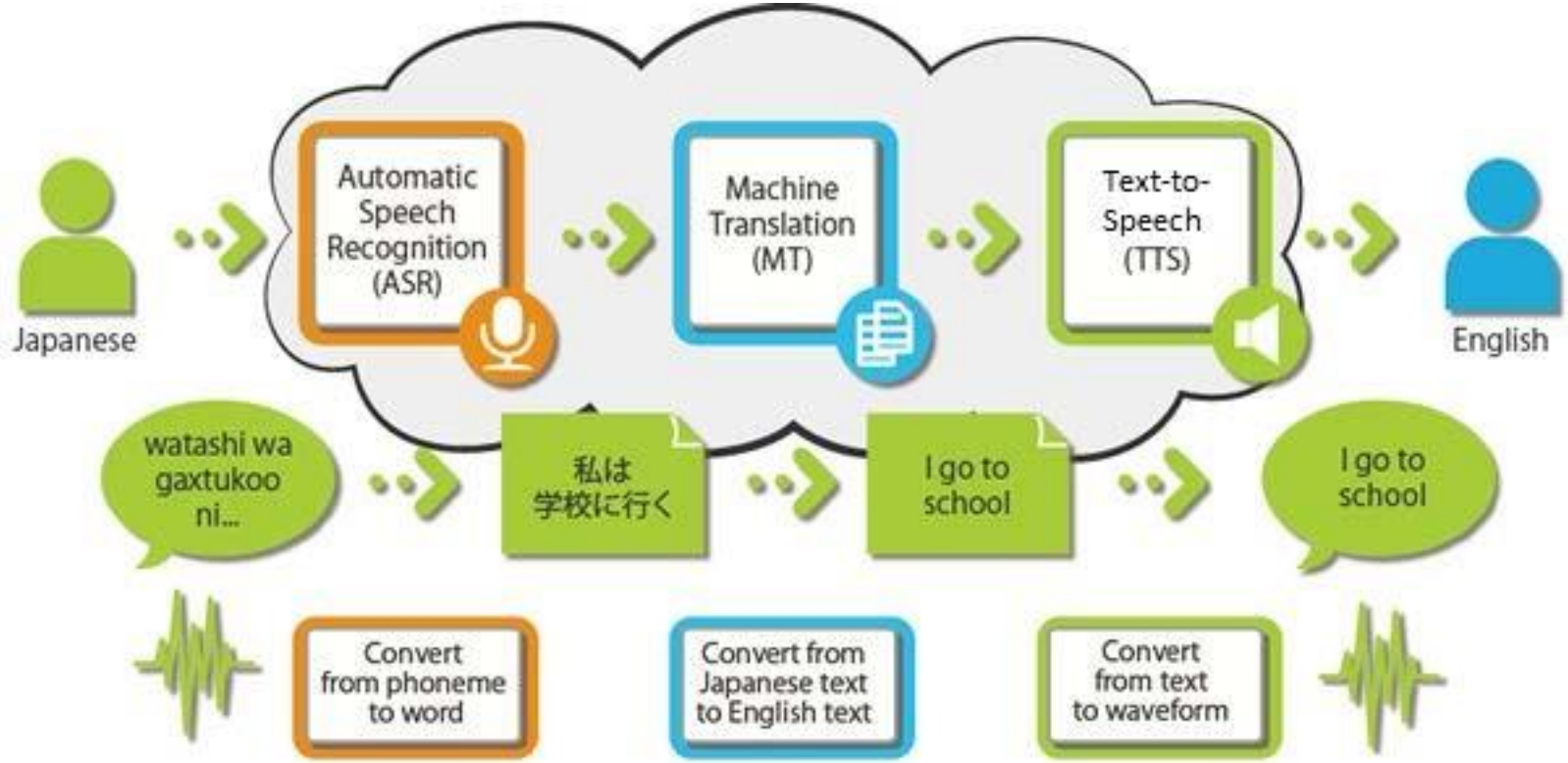
Credit to Andrej Karpathy

Metrics consideration 1

- Are there several metrics?
  - Recommender System
    - Evaluate whether relevant products are ranked at the top of the recommendation list. → Ranking-based Metrics
      - Examples: Precision@K, Recall@K, Mean Average Precision (MAP), and Hit Rate@K.
    - Users are likely to engage with the recommended products (e.g., click or not).→ Classification Metrics
      - Examples: Precision, Recall, F1-score, and related metrics.
- Use the metric closest to your goal but never disregard other metrics.
  - May help identify possible improvements

Metrics consideration 2

- Are there sub-metrics?

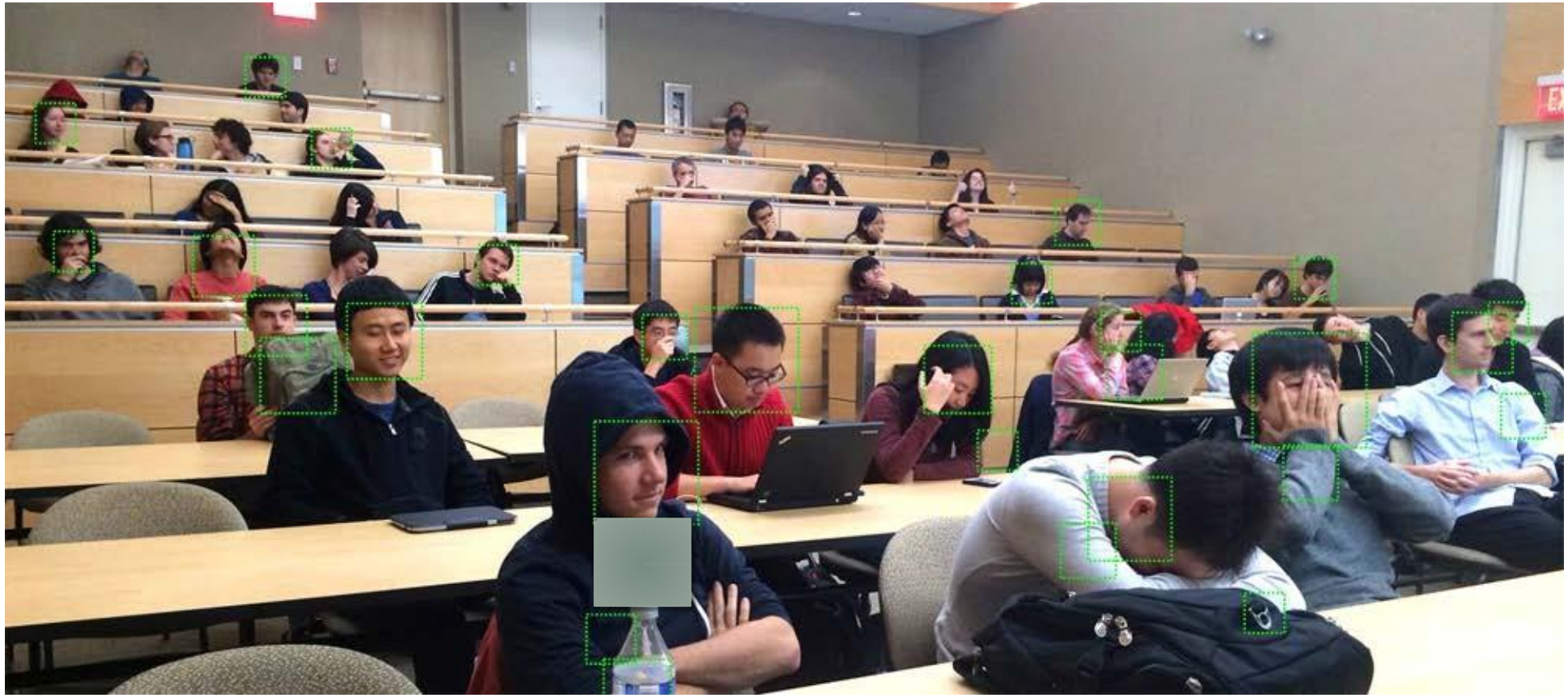


http://www.ustar-consortium.com/qws/slot/u50227/research.html

Metrics definition

- Defining a metric can be tricky when the answer is flexible

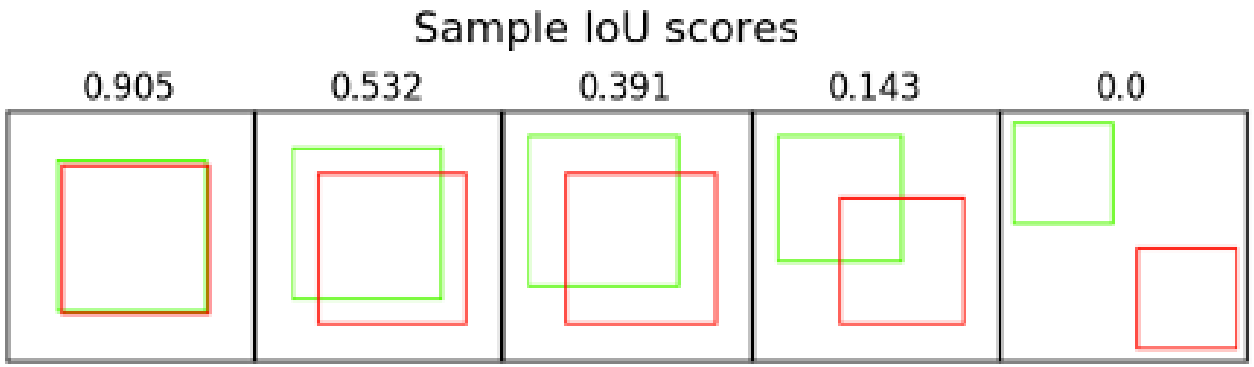
https://www.cc.gatech.edu/~hays/compvision/proj5/



Be clear about your definition of an error before hand!
Make sure that it can be easily calculated!
This will save you a lot of time.

IoU (Intersection over Union)

What IoU score should be considered a detection?



http://ronny.rest/tutorials/module/localization_001/iou/

Commonly used metrics

- Error rate
- Accuracy rate

- Precision
- True positive
- Recall
- False alarm
- F score

A detection problem

- Identify whether an event occur
- A yes/no question
- A binary classifier

Smoke detector

Evaluating a detection problem

- 4 possible scenarios

		Detector	
		Yes	No
Actual	Yes	True positive	False negative (Type II error)
	No	False Alarm (Type I error)	True negative

True positive + False negative = # of actual yes
False alarm + True negative = # of actual no

- False alarm and True positive carries all the information of the performance.

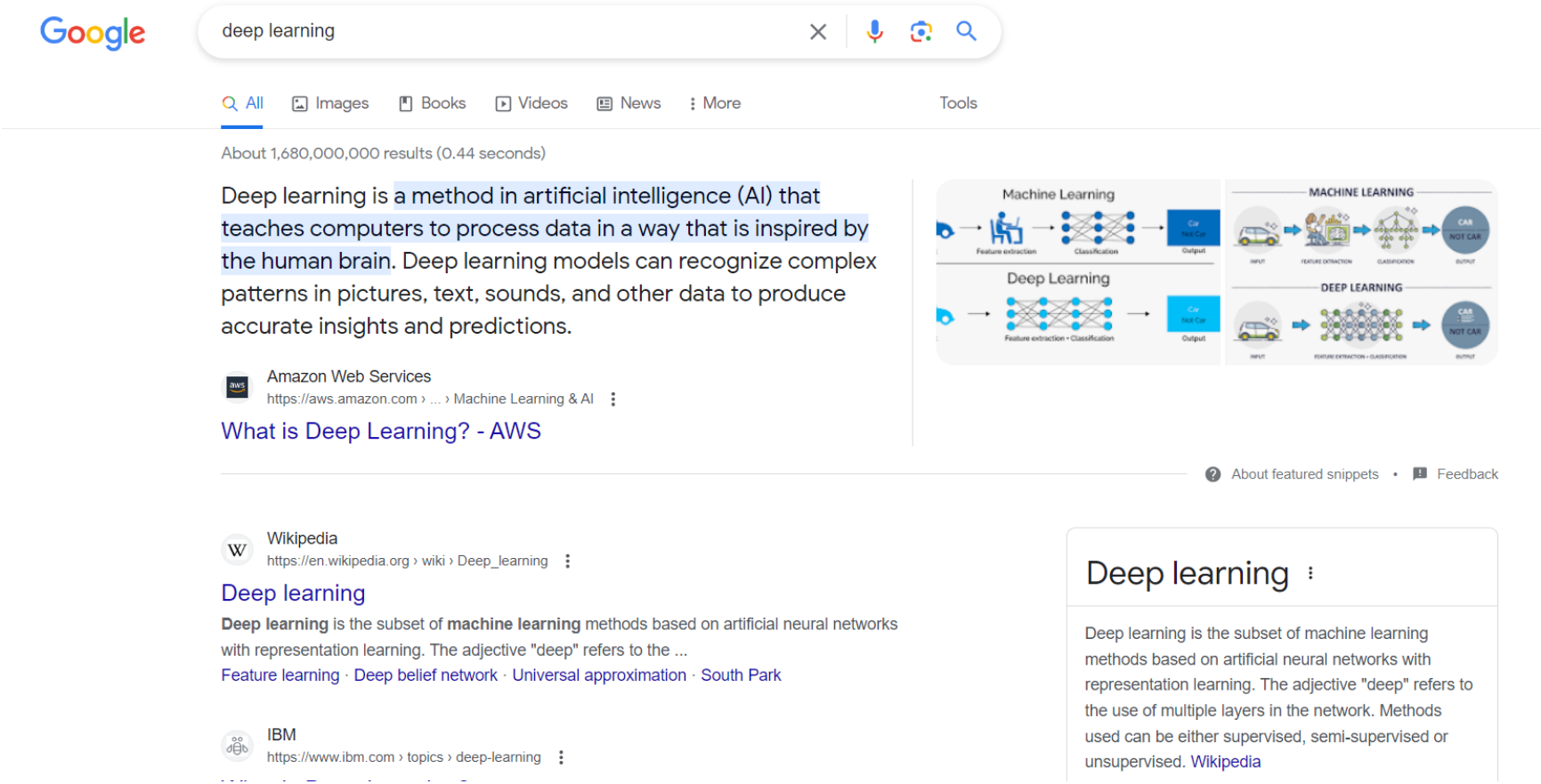
Definitions

		Detector	
		Yes	No
Actual	Yes	True positive	False negative (Type II error)
	No	False Alarm (Type I error)	True negative

- True positive rate (Recall, sensitivity) = # true positive /# of actual yes
- False positive rate (False alarm rate) = # false positive / # of actual no
- False negative rate (Miss rate) = # false negative / # of actual yes
- True negative rate (Specificity) = # true negative / # of actual no

- Precision = # true positive / # of predicted positive

Search engine example



A recall of 50% means?

A precision of 50% means?

Recall/precision

- When do you want high recall?
- When do you want high precision?

- Initial screening for cancer
- Face recognition system for authentication
- Detecting possible suicidal postings on social media
- COVID screening: ATK vs PCR

Usually there's a trade off between precision and recall. We will revisit this later

Let’s consider a case

- A: COVID screening kit 100% recall 10% precision
- B: COVID screening kit 10% accuracy

Let’s consider a case

- A: COVID screening kit 100% recall 10% precision
- B: COVID screening kit 10% accuracy
  - Always answer positive

		Detector	
		Yes	No
Actual	Yes	100	0
	No	900	0

- This population has 10% COVID cases
- What if the population has 20% COVID cases?
  - Prevalence of the data effects the metric values.

Let’s consider another case

- A: no rain predictor has 97% accuracy

Let’s consider another case

- A: no rain predictor has 97% accuracy
  - Always say no rain.
- Accuracy is not a good metric for biased data
- A good model should be better than stupid baselines

		Detector	
		Yes	No
Actual	Yes	0	1
	No	0	30

Definitions 2

- F score (F1 score, f-measure)

$$F_1 = 2 \cdot \frac{1}{\frac{1}{\text{recall}} + \frac{1}{\text{precision}}} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}.$$

  - A single measure that combines both aspects
  - A harmonic mean between precision and recall (an average of rates)

Note that precision and recall says nothing about the true negative

Harmonic mean vs Arithmetic mean

- You travel for half an hour for 60 km/hr, then half an hour for 40 km/hr. What is your average speed?
  - Arithmetic mean = 50 km/hr
  - Harmonic mean

$$\frac{n}{\frac{1}{x_1}+\ldots+\frac{1}{x_n}} = \frac{2}{\frac{1}{40}+\frac{1}{60}} = 48 \text{ km/hr}$$

  - Total distance covered in 1 hour = 30+20 = 50

30 mins	30 mins
60 km/hr	40 km/hr

Harmonic mean vs Arithmetic mean

- You travel for distance X for 60 km/hr, then another X for 40 km/hr. What is your average speed?
  - Arithmetic mean = 50 km/hr
  - Harmonic mean

$$\frac{n}{\frac{1}{x_1}+\ldots+\frac{1}{x_n}} = \frac{2}{\frac{1}{40}+\frac{1}{60}} = 48 \text{ km/hr}$$

  - Total distance covered 2X

Harmonic mean vs Arithmetic mean

- For the arithmetic mean to be valid you need to compared over the same number of hours (denominator)
  - Mean accuracy uses arithmetic mean

- For precision and recall, you have different denominators, but the same numerator, which fits the harmonic mean.

True positive rate (Recall, sensitivity) = # true positive / # of actual yes

Precision = # true positive / # of predicted positive

Evaluating models

- We talked about the training set used to learn the model
- We use a different data set to test the accuracy/error of models – “test set”
- We can still compute the error and accuracy on the training set
- Training error vs Testing error
- We will discuss how we can use these to help guide us later

Other considerations when evaluating models

- Training time
- Testing time
- Memory requirement
- Latency

Topic
Introduction to Pattern Recognition
Break
K-mean and Regression MLE (Maximum likelihood estimate), MAP (Maximum A Posteriori) Bayes Rule and Naive Bayes
Break (No exam for this class)
GMMs (Gaussian mixture models) and EM (Expectation maximization algorithm) Dimensionality Reduction and Visualization Techniques Part01 Dimensionality Reduction and Visualization Techniques Part02
Support Vector Machine (SVM)
Break (No exam for this class) Break (No exam for this class) Break
Basic CNN for computer vision and Basic RNN for sequential data , optimizers (SGD, learning rate setting, momentum, Nesterov momentum and adaptive learning rate)
RNN and CNN in advance (DL architectures (LeNet, AlexNet, ZFNet, VGG, InceptionV1-3, ResNet, Xception, DenseNet, NasNet, EfficientNet, Modern Convolution block))
Basic NLP, Different recurrent architectures (many-to-many, many-to-one, one-to-many, many-to-many (encoder-decoder)), attention, transformers, pre-trained transformer types, Vision transformer
Unsupervised, Semi-Supervised, Self-Supervised, Contrastive Learning
Deep Generative Models (VAE, Diffusion)
Break (No exam for this class) Break (No exam for this class)